

Problem Set 3: Due on Tuesday, February 17, @noon

Search

(100 points total)

1. McCall Search with a Chance of not Receiving a Wage Offer (20 points).

Consider the problem of an unemployed worker who samples wage offers. His wage offer opportunities depend on the state of the economy, $i \in \{\ell, h\}$, which affects the probability p_i with which the unemployed worker receives a wage offer. Let $0 < p_\ell < p_h < 1$. The state is i.i.d. over time, with state i occurring with probability $\pi_i > 0$, where $\pi_\ell + \pi_h = 1$.

In each period, the timing is as follows. In the beginning of each period, the unemployed worker receives unemployment benefit. She then observes the current state of the economy, i . With probability p_i , she receives a wage offer, and with probability $1 - p_i$ she receives no offer. Wage offers are drawn from a cumulative distribution function $F(w)$ on $[0, \bar{w}]$, where F is independent of the state of the economy. Wage offers across periods are i.i.d. If the worker accepts a wage offer w , then she works at w forever starting next period. If the worker rejects a wage offer or receives no offer, then she remains unemployed, and in the next period starts over.

The worker maximizes

$$E \sum_{t=0}^{\infty} \beta^t c_t,$$

where $\beta \in (0, 1)$ and c_t denotes the worker's consumption. If the worker is employed, then $c_t = w$; if the worker is unemployed, then $c_t = b$.

- (a) (7 points) Write down the problem of an unemployed worker in a recursive form.
- (b) (3 points) Prove that the optimal policy of the unemployed worker is of the reservation wage form, that is, there is a unique reservation wage w^* such that the worker accepts all wage offers $w > w^*$ and rejects all wage offers $w < w^*$. Show that this reservation wage does not depend on the state of the economy.
- (c) (15 points) How does w^* change if the probability of the high state, π_h , increases? Prove your answer formally and explain it intuitively.

2. Search with Stochastic Unemployment Benefits (25 points total). This problem asks you to analyze a version of the McCall search model with stochastic and time varying unemployment benefits. The timing is otherwise standard. In each period while the worker is unemployed, the current period's level of unemployment benefit is realized. Then the worker receives a wage offer, drawn from the c.d.f. F on $[0, \bar{w}]$, i.i.d. over time, and decides whether

to accept or reject it. If the worker accepts, he is employed at this wage forever. If he rejects, he receives the unemployment benefit, and enters the next period as unemployed.

The economy can be in one of two states, high or low. When the economy is in the high state, the unemployment benefit equals $\bar{b} < \bar{w}$. When the economy is in the low state, the unemployment benefit equals $\underline{b} < \bar{b}$. When the economy is in the high state, it remains in the high state with probability $p \in (0, 1)$ and switches to the low state with probability $1 - p$. Once the economy is in the low state, it remains in the low state forever.

The worker maximizes the present discounted value of consumption and discounts the future with $\beta \in (0, 1)$.

- (a) (8 points) Write down the recursive problem of the unemployed worker.
- (b) (2 points) Prove that the optimal policy of the unemployed worker is of the reservation wage form.
- (c) (15 points) Derive the expressions for the reservation wages in the two states in terms of the model parameters. What happens to the two reservation wages as $\underline{b}$ increases? Explain your results intuitively.

3. Search with Two Consecutive Wage Offers and Recall (25 points). Consider an unemployed worker searching for a job. In each period while unemployed, the worker receives *two consecutive* wage offers. Specifically, the timing is as follows. First, the worker receives a wage offer w_1 from the *uniform distribution* $U[0, 1]$, which he decides whether to accept or reject. If he rejects, he immediately (i.e., in the *same period*) draws another wage offer, w_2 , from the same distribution. Before making a decision of whether to accept or reject it, he learns whether the previous wage offer w_1 is still available. That is, the worker might have a chance to *recall* the previous job offer. The old offer is still available with probability p , in which case the worker decides which of the two wage offers— w_1 or w_2 —to accept, if any. With probability $1 - p$ recall is not possible, and the worker decides whether to accept w_2 or reject it. Whenever the worker accepts a wage offer, he will be forever employed at that wage starting from the next period. The unemployed worker receives an unemployment benefit $b \in [0, 1)$ in the beginning of each period. The worker maximizes the expected present discounted value of his income stream and discounts the future with $\beta \in (0, 1)$.

- (a) (10 points) Write down the unemployed worker's problem in a recursive form.
- (b) (10 points) Carefully describe the optimal strategies of the unemployed worker. For any cutoffs (reservation wages) in the optimal strategies that you describe, determine how these cutoffs compare to each other. Does the worker ever recall the previous wage offer? If so, under what conditions? Explain your answer intuitively.
- (c) (5 points) Now compare this model to the basic McCall's model, where there is a single offer each period. Specifically, in that model, if the worker had an option to recall the offer from the previous period, when would he do it, if ever? Explain your answer

intuitively, and explain how it compares to your findings in the above model. Which feature of the model(s) drives this comparison? Explain.

4. **Search with Temporary Jobs (25 points total).** Consider a risk-neutral worker searching for a job. He maximizes the expected present discounted value of his income stream and discounts the future with $\beta \in (0, 1)$. In each period, the unemployed worker receives one job offer with probability π_s that depends on the state of the economy s , which can be high (h) or low (ℓ). Offers arrive more frequently in high vs low state of economy, i.e., $0 < \pi_\ell < \pi_h < 1$. Wage offers are drawn from the probability distribution with c.d.f. $F(w)$ on $[0, \bar{w}]$, independent of the state s .

All jobs employ for one period only. The timing is as follows. The state of the economy is realized. A worker who is currently unemployed receives unemployment insurance benefit b in the beginning of a period, and then draws a job offer (he will not receive one with probability $1 - \pi_s$). If he rejects, he remains unemployed and searches again next period. If he accepts the job with wage w , he works the next period at that wage, and becomes unemployed again at the end of the next period. (So a worker who a job in period t will spend period $t + 1$ working, and will search again in period $t + 2$.)

Suppose first that the state of the economy is i.i.d. over time, and the high state occurs with probability $p \in (0, 1)$. Denote the value to the unemployed worker at the beginning of a period if the state is $s \in \{h, \ell\}$ by U_s .

- (a) (8 points) Write the problem of an unemployed worker recursively. Describe his optimal strategy. How do the reservation wages compare depending on the state of the economy? Carefully explain intuition behind your answer.

Now suppose that the state of the economy follows a Markov process, with the probability of staying in each state (persistence) equal to $p > 1/2$.

- (b) (12 points) Write the recursive problem of an unemployed worker in this case. Describe his optimal strategy. Conjecture that $U_h > U_\ell$. Given this conjecture, how do reservation wages compare depending on the state of the economy? Carefully explain intuition behind your answer.
- (c) (5 points) Sketch how you would prove that indeed $U_h > U_\ell$.